

Protein Homology Detection

Part 2: Multiple Sequence Methods

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Multiple Sequence Methods

- Although pairwise sequence comparison is the single most widely used similarity/homology detection method, multiple sequence methods are much more powerful
- Allows a statistical model to be built of a whole family of sequences rather than just an isolated pair of sequences

```

HIV-1      PQITLW--QRPLVTIKIG-----GQLKEALLDTGADDTVLEE--M--NL--PGKWK--PK
HIV-2      PQFSLW--KRPVVTAHIE-----GQPVEVLDTGADDSIVAG--I--EL--GNNYS--PK
EIAV       VTYNLE--KRPTTIVLIN-----DTPLNVLLDTGADTSVLTTHYNNRLKYRGRKYQ--GT
FIV        YNKVGTTTTLE--KRPEILIFVN-----GYPIKFLDTGADITILNRRDF--QVKNSIENG--RQ
MAV        LAMTMEHKDRPLVRVILTNTGSHPVKQRSVYITALLDSGADITIISEEDW-PADW-P-VMEAANP
HTLV-1     PVIPLDPARRPVKAQIDTQT-----SHPKTIEALLDTGADMTVLPPIA-L--FSSNT-PLK--NT
BLV        LSIPLA-RSRPSVAVYLSGPWL---QPSNQALMLVDTGAENTVLPQN-W--LVRDY-PRI--PA
MMLV       TLDDQGGQGGQDPP--PEPRITLVKG-----GQPVTFLVDTGAQHSVLTQ--N--P--GPLSDK--SA
WDSV       PIDCPYEKSGTKTTQDVITTKNAEIMVTVN-----HTKIPMLVDTGACLTAIIGGAATVVP-DLKLNT--EI
MPMV       WVQPITC---QKPSLTLWLD-----DKMFTGLIDTGADVTIKLEDW-PPNW--PITDT-LT
MMTV       WVQEISD---SRPMLHIYLN-----GRRFLGLLDTGADKTCIAGRWD-PANW--PIHQT-ES
HFV        MNPLQ---LQPLPAEIK-----GTKLLAHWDSGATTICIPESFLEDEQP----IK--KT
  
```

*Example of a simple
multiple sequence
alignment*

Viruses (2010) 2:147-165

```

HIV-1      MIGGI GGGFI--KVRQY-DQIPVEIC-----GH--KAIG-TVLVG---PTPVNIIGRNLLTQIGCTLNF
HIV-2      IVGGI GGGFI--NTKY--KNVEIEVL-----NK--KVRA-TIMTG---DTPINIIGRNILTALGMSLNL
EIAV       GIGGV GGNV--ETFS-T-P-VTIKKK-----GR--HIKT-RMLVA---DIPVTILGRDILQDLGAKLVL
FIV        NMIGV GGGK--RGTNY--INVHLEIRDENYK-TQ--CIFG-NVCVLEDNSLIQPLLRDRNMIKFNIRLVM
MAV        QIHGI GGGI--PMKSRDMIEVGVINRDGSLER--PLLL-FPAVA---MVRGSIILGRDCLQGLGLRLTNL
HTLV-1     SVLGA GGGTQDHFKL-TSLPVLIRLP-----FRTTPIVLTSC-LVDTKNN--WAIIGRDALQQCQGVLYLPEAKRPPVIL
BLV        AVLGA GGVSRNRYNWLQGGPLTLALK-----PEGPFITIPKILVDTFDK--WQILGRDVL SRLQASISIPEEVRPPMVG
MMLV       WVQGA TGKK--RYRW-TDRKVHLA-----TG--KVTH-SFLHV--PDCPYLLGRDLLTKLQAIHFEGSGAQVMGPMGQPLQVL
WDSV       IAVGI SAEP--PHVLAKPTKIQIE-----NT--NIDI-SPWYN--PDQTFHILGRDRLSKMRAIVSEKNGEMTVLLPPTYHKQL
MPMV       NLRGI GQSN--NPKQSSKYLTRWDKEN---NS--GLIK-PFVI---PNLPVNLWGRDLLSQMKIMMCS
MMTV       SLQGL GMAC--GVARSQPLRWQH-ED---KS--GIIH-PFVI---PTLPFTLWGRDIMKDIKVRMLTDSPPDSQDL
HFV        LIKTI HGEK--QONVY--YVTFKVK-----GR--KVEA-EVIA---SPYEYILLSPTDVP--WLTQQPLQLTILVPLQEYQEKILS
  
```

MSAs and machine learning

- A multiple sequence alignment provides a concise representation of the entire evolutionary history of a protein family
- It can identify regions which are conserved and thus likely to be functionally or structurally important
- A complication for ML is that not only can proteins have different lengths, families can be of any size from 1 to 100,000 or more members
- How can we represent this data like this so that it can be used as an input to a ML model?

Making Multiple Sequence Alignments

- By hand/eye
 - Only possible for a few sequences
 - Lacks objectivity and repeatability
- Rigorous dynamic programming
 - Can be extended from pairwise alignment to alignment of any number of sequences
 - VERY slow
- Heuristics
 - Build up multiple sequence alignment from pairwise alignments

Typical Heuristic Approach to Multiple Sequence Alignment

- Compare all sequences pairwise (e.g. using NW algorithm). For n sequences there are $\frac{n(n-1)}{2}$ unique pairs.
- Construct a *guide tree* based on these alignments.
 - This aims just to roughly approximate the true phylogeny of sequences
- Build the multiple alignment by following the order of the guide tree branches – i.e. first aligning the most similar pair of sequences within each cluster, then the next most similar pair and so on, until all sequences have been aligned.
- This strategy serves to minimize the number of gaps required to complete the whole alignment

Example of Multiple Sequence Alignment

>O17208

KYME LKCGALVARLHTELFICPGIREKCI EVLDCPGELLTPVEFTIKA ESKQKDWKGAIKHNGRMLRYASCSFQLLKTEQFNRTLMEFKQLD

>LY10_HUMAN

PLL PVT CGGVKGILHKKKLQQGILVKCIQTEDGKWFTPTFEFEIKGGHARSKNWRLSVRCGGWPLRWLMENGFLP

>O75450

SELPVTCGEVKG TLYKERFKQGT SKKCIQSEDKKWFTPREFEIEGDRGASKNWKLSIRCGGYTLKVL MENKFLP

>Q14976

SKSPVTCGEAKGILYKKKMKHGS SVKCI RNEDGTWLT PNEFEVEGKGRNAKNWKRNI RCEGMTLGELLKSGLLL

>O75511

SELPVRCRNISGTLYKNKLGSGGRGRCIKQGENWYSPTEFEAMAGRASSKDWKRSIRYAGRPLQCLIQDGILN

>O35893

AELPAVCGNVQGFLNKEKFKQGIYVRSIRSETGRLFTPMDFEIEGNCEKAKNWRQTIRCKGWTREL IQKGV LQ

>Q18624

TTVPVTCGYVKGIMHLKLFRCPGIRQACIEYESQYLTPKQFTELGGKGRQKDWKACIRVENVSLRTLLENKTID

>Q24180

EVFQIRCKTTCAELYRSKLGSGGRGRCVKYKDKWHTPSEFEHVCGRGSSKDWKRSIKYGGKSLQSLIDEGLT

>Q18171

PTVPVSCGVVNGKMHLNLFMCPIHQPCIEVGNDLLSPKQFTIRGDKERQKDWKASIRVGRSSLRTHMEAMTID

>Q93779

IIFQVKCGTLTAKMVTKLFTCPGIHQNCILIDGETEYISPKEFTVRANKDKQKDWKGSIRIGKSNLRTL MEMRSLD

>O88873

IVYPITCGDSRANLIWRKFVCPGINVKCVQYDEHVISPKEFVHLAGKSTLKDWKRAIRMNGIMLRKIMDSGELD

Input
Sequences



Multiple Sequence Alignment by CLUSTALW

CLUSTALW

MAFFT

PRRN

[Help](#)

General Setting Parameters:

Output Format: Pairwise Alignment: ☒ FAST/APPROXIMATE ☐ SLOW/ACCURATEEnter your [sequences](#) (with labels) below (copy & paste): ☒ PROTEIN ☐ DNA

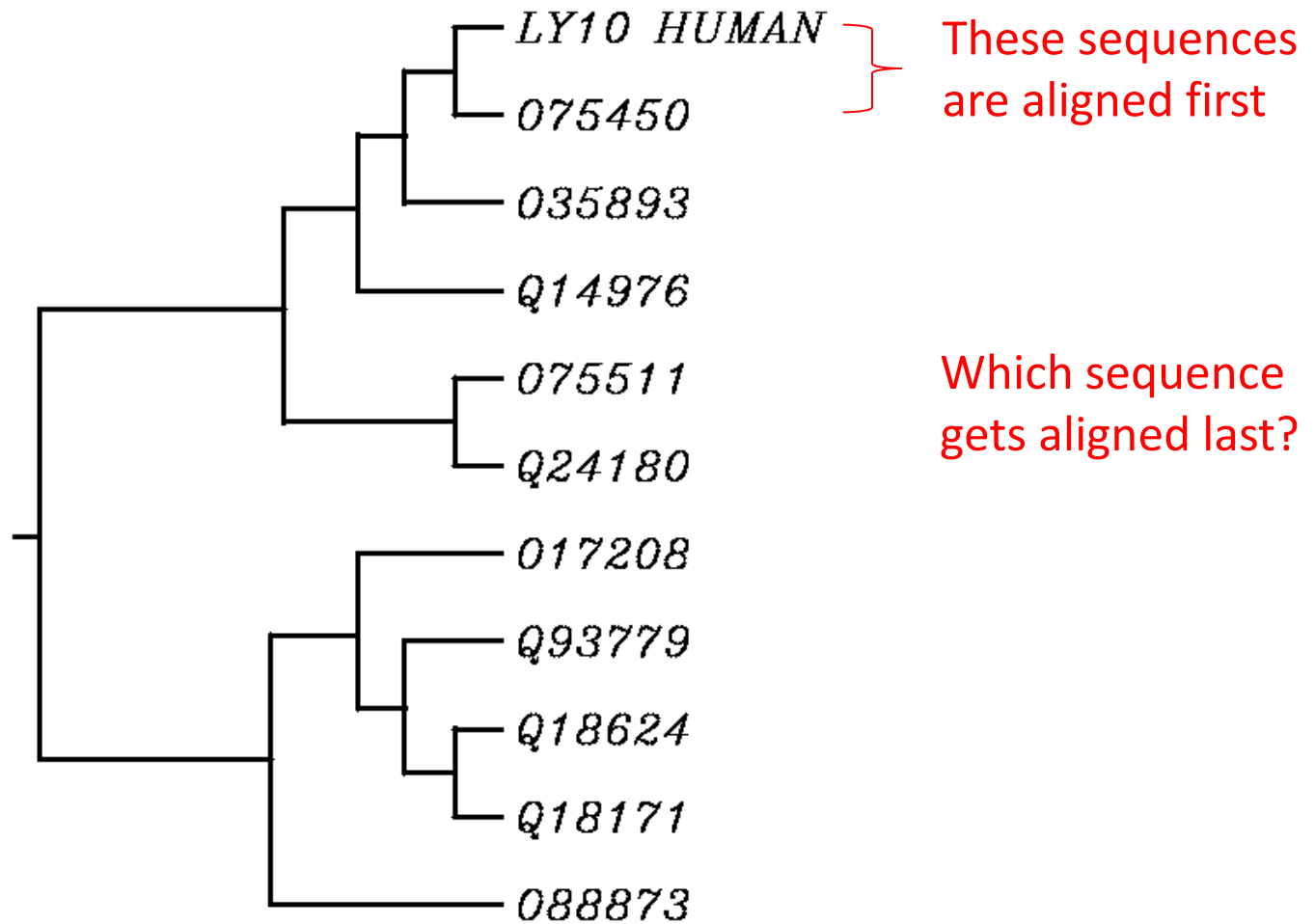
Support Formats: FASTA (Pearson), NBRF/PIR, EMBL/Swiss Prot, GDE, CLUSTAL, and GCG/MSF

```
>O17208
KYMELKCGALVARLHTELFICPGIREKCIEVLDCPGELLTPVEFTIKAESKQKDWKGAI
KHNGRMLRYASCSFQLLKTEQFNRTLMEFKQLD
>LY10_HUMAN
PLLPVTCGGVKGILHKKKLQQGILVKCIQTEDGKWFTPTFEIKGGHARSKNWRLSVRCG
GWPLRWLMENGFLP
```

Or give the file name containing your query

No file selected.

UPGMA Guide Tree



LY10_HUMAN	1	PLLPVTCGGVKGILHKKKLQQ-GILVKCIQTEDGK--WFTPTFEFEIKGGHARSKNWRLSV
O75450	1	SELPVTCGEVKGTLYKERFKQ-GTSKKCIQSEDKK--WFTPREFEIEGDRGASKNWKLSI
O35893	1	AELPAVCGNVQGFLNKEKFKQ-GIYVRSIRSETGR--LFTPMDFEIEGNCEKAKNWQRQTI
Q14976	1	SKSPVTCGEAKGILYKKKMKH-GSSVKCIRNEDGT--WLTPNEFEVEGKGRNAKNWKRNI
O75511	1	SELPVRCRNISGTLYKNKLGS-GGRGRCIKQGE-N--WYSPTEFEAMAGRASSKDWKRSI
Q24180	1	EVFQIRCKTTCAELYRSKLGS-GGRGRCVKYKD-K--WHTPSEFEHVCGRGSSKDWKRSI
O17208	1	KYMELKCGALVARLHTELFICPGIREKCIENVLDCPGELLTPVEFTIKAEKSKQKDWKGAI
Q93779	1	IIFQVKCGTLTAKMVTKLFTCPGIHQNCILIDGET-EYISPKEFTVRANKDKQKDWKGS
Q18624	1	TTVPVTCGYVKGIMHLKLFRCPGIRQACIEYESQY---LTPKQFTELGGKGRQKDWKACI
Q18171	1	PTVPVSCGVVNGKMHNLNFMCPGIHQPCIEVGNDL---LSPKQFTIRGDKERQKDWKASI
O88873	1	IVYPITCGDSRANLIWRKFVCPGINVKCVQYDEHV---ISPKEFVHLAGKSTLKDWKRAI
consensus	1	lpvtCg v g l k kf Gir kci ed wltP eFei g h r KdWk si

LY10_HUMAN	58	RCG-----GWPLRWLMENGFLP
O75450	58	RCG-----GYTLKVLNENKFLP
O35893	58	RCK-----GWTLRELIQKGVLP
Q14976	58	RCE-----GMTLGELLKSGLLL
O75511	57	RYA-----GRPLQCLIQDGILN
Q24180	57	KYG-----GKSLQSLIDEGTLT
O17208	61	KHNGRMLRYASCSFQLLKTEQFNRTLMEFKQLD
Q93779	60	RIG-----KSNLRTLMEMRSLD
Q18624	58	RVE-----NVSLRTLLENKTID
Q18171	58	RVG-----RSSLRTHMEAMTID
O88873	58	RMN-----GIMLRKIMDSGELD
consensus	61	r g g tlr lme g l

Protein Sequence Patterns

- Regular Expressions (somewhat obsolete)
 - E.g. PROSITE
 - Describes specific residue patterns which must be present
 - Strict YES/NO rules
- Profiles
 - E.g. Pfam
 - Describes features in terms of probabilities
 - Matches can be weak or strong

Sequence Profiles

- Sequence profiles provide a more general alternative to regular expression patterns
- Can describe each column of a multiple sequence alignment by the fraction of each amino acid observed
- These fractions can be treated as probability estimates
- For example ...

Observed residues at given site: AAAAAGGSSS

$$p(A) = 0.5, p(G) = 0.2, p(S) = 0.3$$

$$p(\text{RNDCQEHILKMFPTWYV}) = 0.0$$

Simple profile example

Alignment:

Position 12345
Seq1 CCGTL
Seq2 CGHSV
Seq3 GCGSL
Seq4 CGGTL
Seq5 CCGSS

	1	2	3	4	5
C	0.8	0.6	0	0	0
G	0.2	0.4	0.8	0	0
H	0	0	0.2	0	0
S	0	0	0	0.6	0.2
T	0	0	0	0.4	0
L	0	0	0	0	0.6
V	0	0	0	0	0.2

⋮

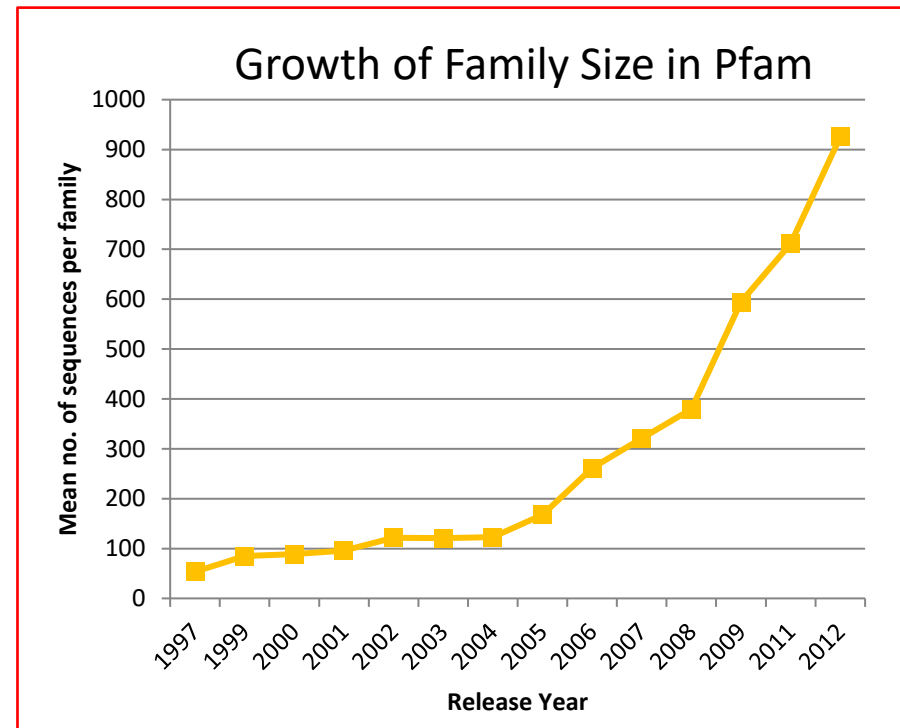
Row for each amino acid (20 total)

Using sequence profiles

- We can convert the frequencies in the sequence profile into log likelihood ratios by dividing each frequency by the background frequency observed in the sequence database and taking logs
- These values can be used as scores for calculating a sequence alignment in place of the LLRs in the 20x20 score matrix (e.g. BLOSUM)
- This gives profiles their alternative name: **Position-Specific Score Matrices (PSSMs)**

Pfam

- Protein **families** database of alignments and HMMs
- Uses profile-HMMs to represent families.
- For each family in Pfam you can:
 - Look at multiple alignments
 - View protein domain architectures
 - Examine species distribution
 - Follow links to other databases
 - View known protein structures



Pfam web site



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Pfam 27.0 (March 2013, 14831 families)

The Pfam database is a large collection of protein families, each represented by **multiple sequence alignments** and **hidden Markov models (HMMs)**. [More...](#)

QUICK LINKS

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Look at the domain organisation of a protein sequence
Find the domains on a PDB structure
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Or view the [help](#) pages for more information

Family: SH2 (PF00017)

379 architectures

9081 sequences

8 interactions

283 species

390 structures

Summary

Domain organisation

Clan

Alignments

HMM logo

Trees

Curation & model

Species

Interactions

Structures

Jump to...

enter ID/acc Go

Summary: SH2 domain

Pfam includes annotations and additional family information from a range of different sources. These sources can be accessed via the tabs below.

Wikipedia: SH2 domain

Pfam

InterPro

This is the Wikipedia entry entitled "[SH2 domain](#)". [More...](#)

SH2 domain

[Edit Wikipedia article](#)

The **SH2** (Src Homology 2) **domain** is a structurally conserved [protein domain](#) contained within the [Src oncoprotein](#)^[2] and in many other [intracellular signal-transducing](#) proteins.^[3] SH2 domains allow proteins containing those domains to dock to phosphorylated tyrosine residues on other proteins. SH2 domains are commonly found in adapter proteins that aid in the signal transduction of receptor tyrosine kinase pathways.^[4]

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- Introduction
- Binding and phosphorylation
- Diversity
- Function
- Examples
- References
- External links

Introduction[\[edit\]](#)

Protein-protein interactions play a major role in cellular growth and development. Modular domains, which are the subunits of a protein, moderate these protein interactions by identifying short peptide sequences. These peptide sequences determine the binding partners of each protein. One of the more prominent domains is the SH2 domain. SH2 domains play a vital role in cellular communication. Its length is approximately 100 amino acids long and it is found within 111 human proteins.^[5] Regarding its structure, it contains 2 [alpha helices](#) and 7 [beta strands](#). Research has shown that it has a high affinity to [phosphorylated tyrosine](#) residues and it is known to identify a sequence of 3-6 amino acids within a peptide [motif](#).

Binding and phosphorylation[\[edit\]](#)

SH2 domains typically bind a [phosphorylated tyrosine residue](#) in the context of a longer peptide motif within a target protein, and SH2 domains represent the largest class of known pTyr-recognition domains.^{[6][7]}

Phosphorylation of tyrosine residues in a protein occurs during signal transduction and is carried out by [tyrosine kinases](#). In this way, phosphorylation of a [substrate](#) by tyrosine kinases acts as a switch to trigger binding to an SH2 domain-containing protein. Many tyrosine containing short linear motifs that bind to SH2 domains are conserved across a wide variety of higher Eukaryotes.^[8] The intimate relationship between tyrosine kinases and SH2 domains is supported by their coordinate emergence during eukaryotic evolution.

Diversity[\[edit\]](#)

SH2 domains are not present in [yeast](#) and appear at the boundary between [protozoa](#) and [animalia](#) in organisms such as the social amoeba *Dictyostelium discoideum*.^[9]

A detailed [bioinformatic](#) examination of SH2 domains of [human](#) and [mouse](#) reveals 120 SH2 domains contained within 115 proteins encoded by the human genome,^[10] representing a rapid rate of evolutionary expansion among the SH2 domains.

A large number of SH2 domain structures have been solved and many SH2 proteins have been knocked out in mice. Information generated on the Mouse Knockouts can be found on the sh2.uchicago.edu website.^[11]

Function[\[edit\]](#)

The function of SH2 domains is to specifically recognize the phosphorylated state of tyrosine residues, thereby allowing SH2 domain-containing proteins to localize to tyrosine-phosphorylated sites. This process constitutes the fundamental event of signal transduction through a membrane, in which a signal in the extracellular compartment is "sensed" by a receptor and is converted in the intracellular compartment to a different chemical form, i.e. that of a phosphorylated tyrosine. Tyrosine phosphorylation leads to activation of a cascade of protein-protein interactions whereby SH2 domain-containing proteins are recruited to tyrosine-phosphorylated sites. This process initiates a series of events which eventually result in altered patterns of gene expression or other cellular responses. The SH2 domain, which was first identified in the oncoproteins Src and Fps, is about 100 amino-acid residues long. It functions as a regulatory module of intracellular signaling cascades by interacting with high affinity to phosphotyrosine-containing target peptides in a sequence-specific and strictly phosphorylation-dependent manner.



Crystallographic structure of the SH2 domain. The structure consists of a large [beta sheet](#) (green) flanked by two [alpha-helices](#) (orange and blue).^[1]

Identifiers

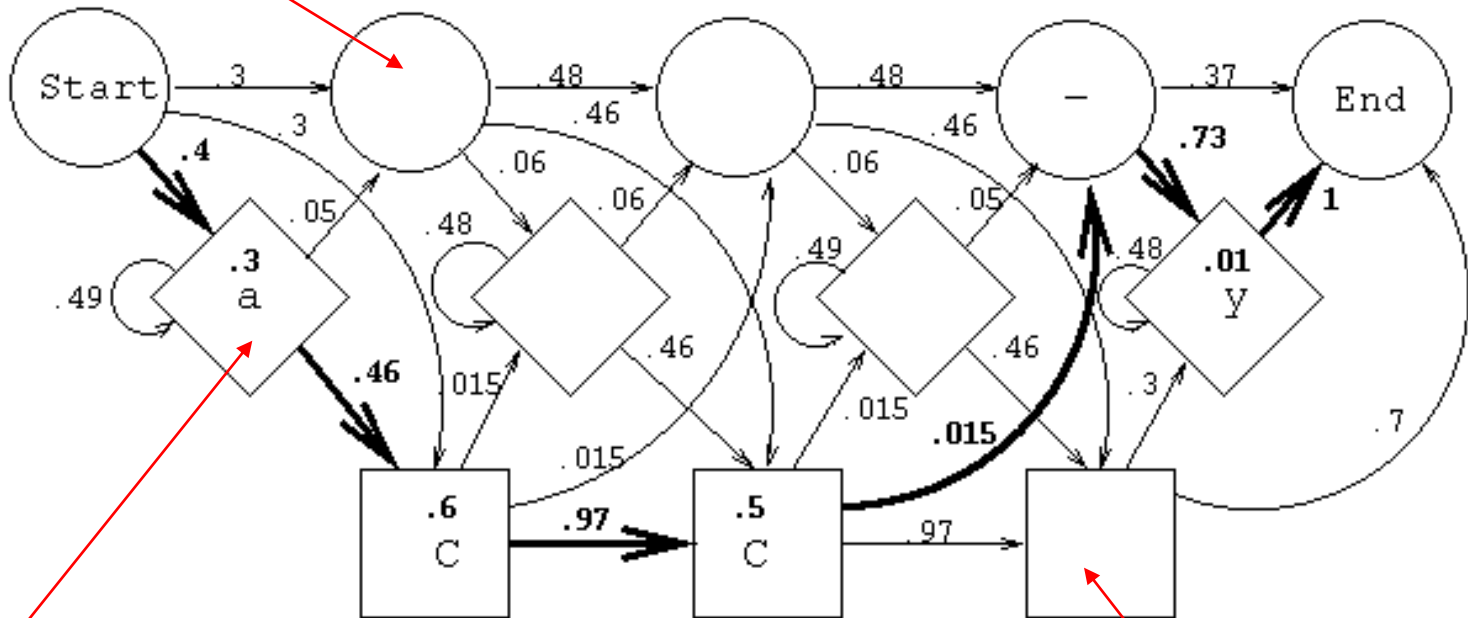
Symbol	SH2
Pfam	PF00017 ↗
InterPro	IPR000980 ↗
SMART	SH2 ↗
PROSITE	PD0C50001 ↗
SCOP	1sha ↗
SUPERFAMILY	1sha ↗
OPM protein	1xa6 ↗
CDD	cd00173 ↗

Available protein structures: [\[show\]](#)

Matching Sequences with HMMs

Possible HMM for sequence ACCY

Delete states

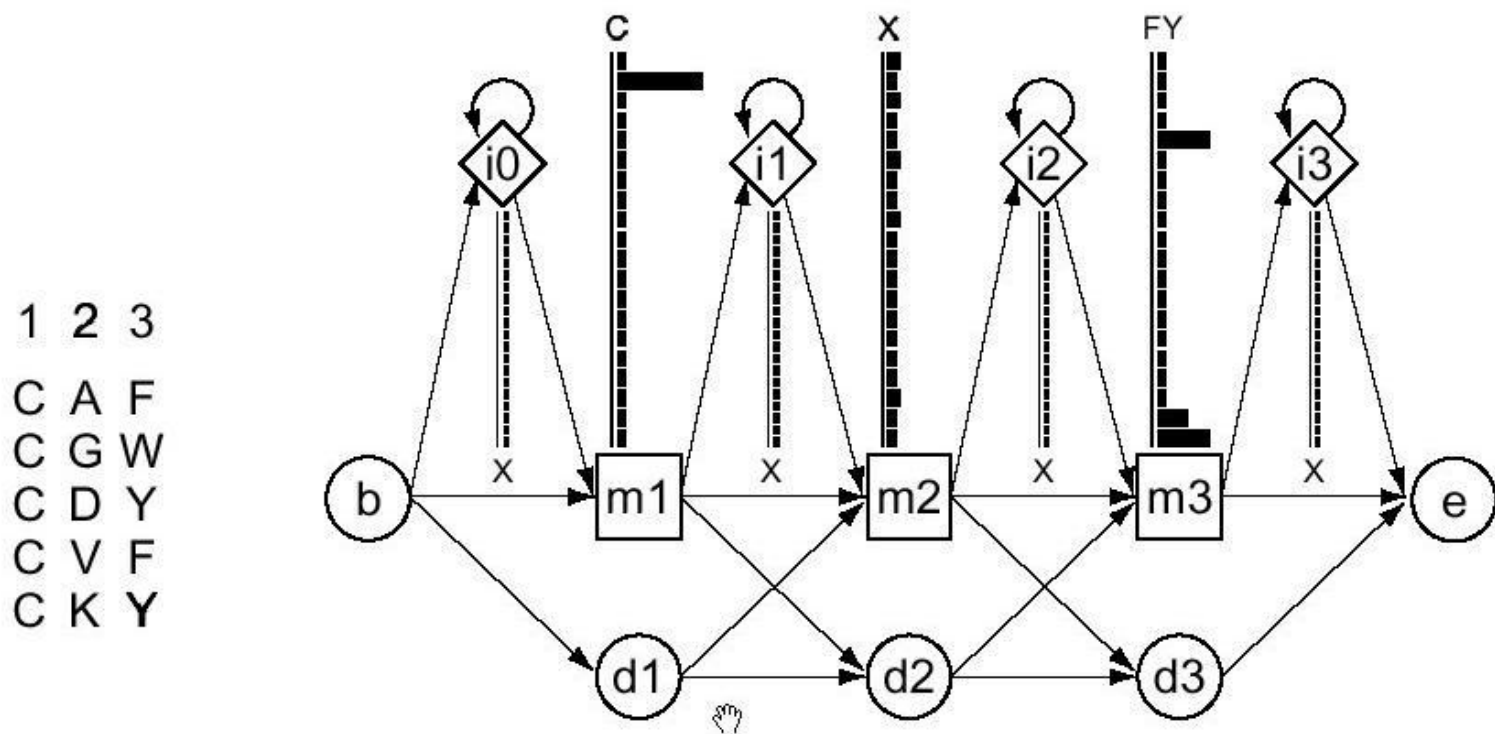


Insert states

Match states

Matching Sequences with Profile HMMs

Example for a trivial alignment

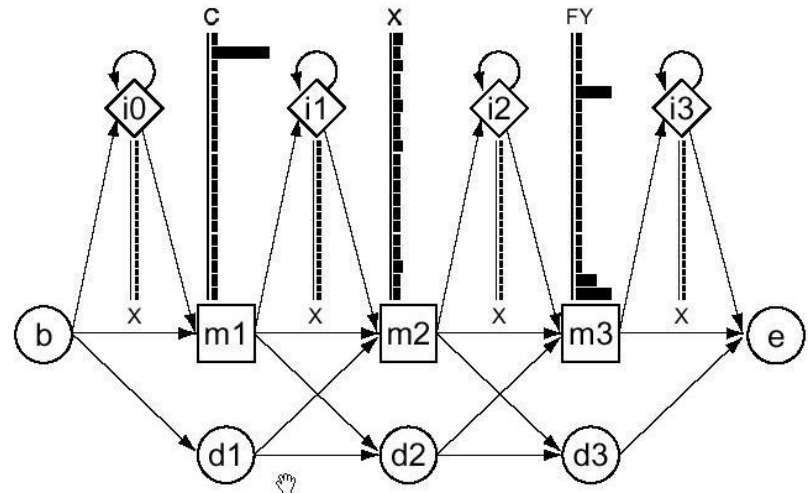


Matching Sequences with HMMs

Example for a trivial alignment

What is the chance
that the sequence
ASCGGFPGG
was generated by
this HMM?

	1	2	3
C	A	F	
C	G	W	
C	D	Y	
C	V	F	
C	K	Y	



Most likely: b i0 i0 m1 m2 i2 m3 i3 i3 i3 e

Much less likely: b i0 i0 m1 d2 m3 i3 i3 i3 i3 e

The Three Main HMM Algorithms again

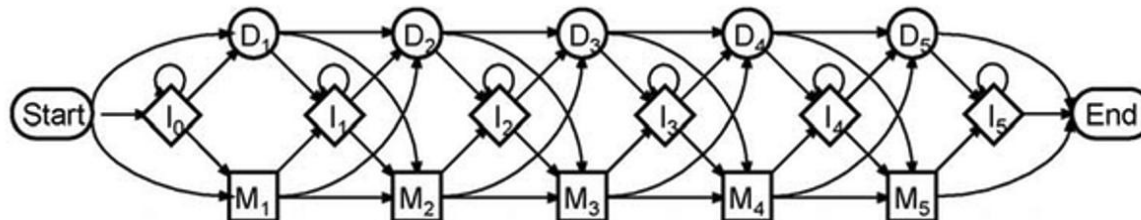
1. The **Viterbi** algorithm: get the most probable state sequence i.e. alignment
2. The **Forward** algorithm: get the probability of each state at each position and the overall match probability
3. **Expectation Maximization**: derive the parameters of the model from the data

These algorithms can be applied to any HMM applications, only the design of the model (i.e. state machine) will be different.

Simplified View of a Profile HMM's States

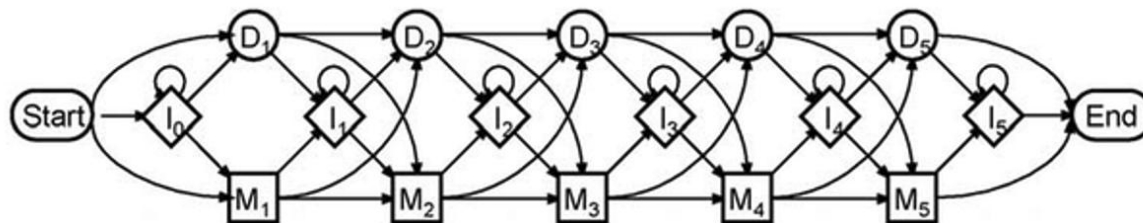
A profile HMM has three sets of states:

- Match states: $M_1, \dots, M_n$
- Insert states: $I_1, \dots, I_n$
- Delete states: $D_1, \dots, D_n$



Profile HMM Viterbi Algorithm

- Define $v_j^M(i)$ as the (log) likelihood score of the best path for matching sequence $x_1..x_i$ to the HMM, ending with x_i being emitted by match state M_j .
- Define $v_j^I(i)$ and $v_j^D(i)$ similarly for insert and delete states.



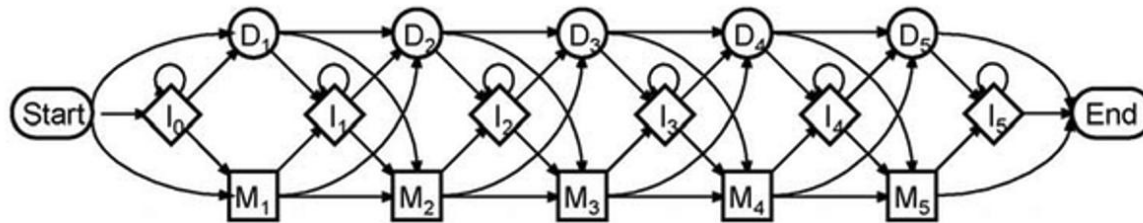
Profile HMM Viterbi Algorithm (for the Match States)

Observed match state
emission probability

Transition
probabilities

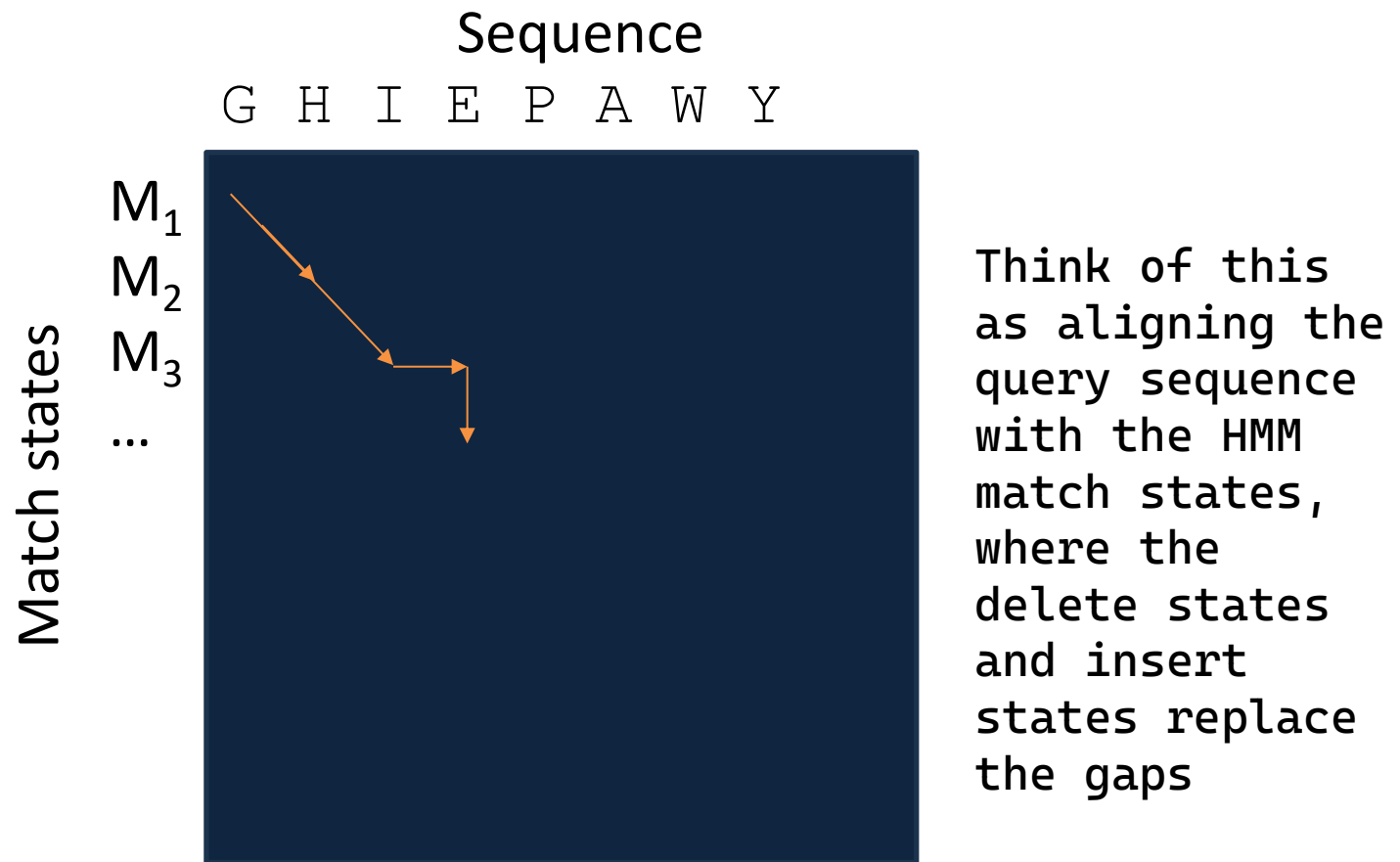
$$V_j^M(i) = \log \frac{e_{M_j}(x_i)}{q_{x_i}} + \max \begin{cases} V_{j-1}^M(i-1) + \log a_{M_{j-1}M_j}, \\ V_{j-1}^I(i-1) + \log a_{I_{j-1}M_j}, \\ V_{j-1}^D(i-1) + \log a_{D_{j-1}M_j}; \end{cases}$$

Background
probability



See p.109 onwards of “Biological sequence analysis” book by Durbin et. al. for the insert and delete state formulae plus further implementational details.

Viterbi is equivalent to dynamic programming sequence alignment



Horizontal moves would require using an insert state, vertical moves would use up a delete state

Practical Issues

- In practice, HMMs are again implemented with **log-likelihood scores** rather than probabilities
 - **Allows probability products to be replaced by sums** & allows the same statistical models used by e.g. BLAST to be used to compute E-values etc.
- Calculations usually done with integer arithmetic for speed
 - LLR scores scaled by e.g. 1000 and truncated to integers

Training HMMs

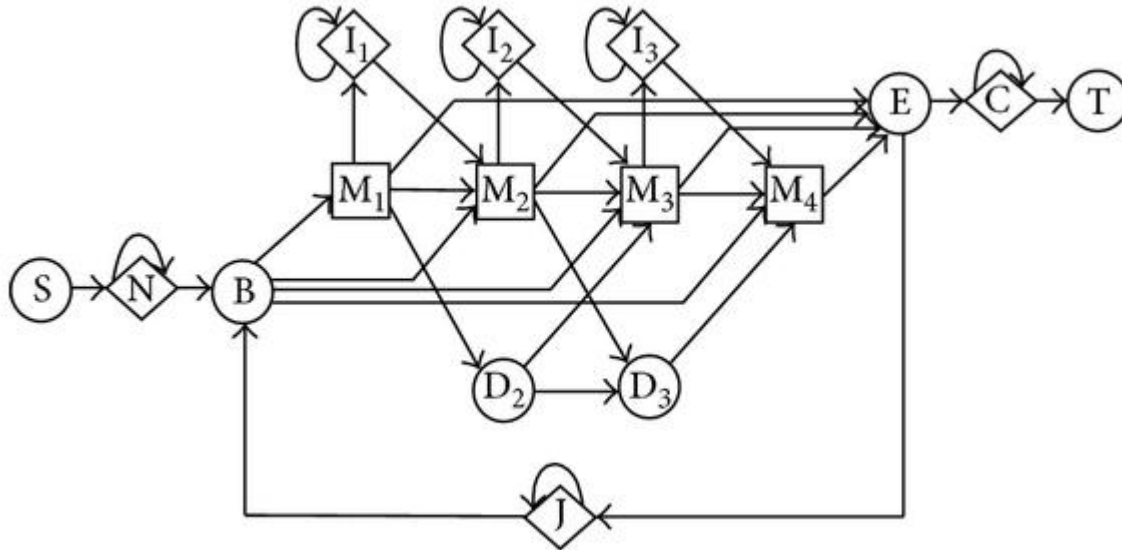
- Start with aligned sequences (training set)
- Baum-Welch Expectation Maximisation
 - Iteratively change all parameters to maximise total expectation score for training sequences
 - Slow but fairly accurate
- Viterbi training
 - Computes new parameters according to only most probable path and score
 - Fast but approximate
- Annealing
 - Avoids local minima
 - Very slow, very accurate, but non-deterministic... and did I mention slow?

HMMER

- HMMER is now the most widely used tool for building and using protein HMMs
- Uses a slight variation of the state model called “Plan 7” (with some extra features)
- Now includes several heuristic speedups to allow searching of very large databases

See <http://hmmer.org>

HMMER Plan 7 Model



<http://dx.doi.org/10.1155/2013/252183>

Compare this to the classic profile HMM state diagram shown earlier. Note the absence of I→D and D→I transitions. Also note the addition of N, C and J states and transitions that allow the intended mode of alignment (local/global/repeat) to be stored in the model.

HMMER Plan 7 File Example

```
HMMER2.0
NAME   rrm
ACC    PF00076
DESC   RNA recognition motif. (aka RRM, RBD, or RNP domain)
LENG   72
ALPH   Amino
RF      no
CS      no
MAP     yes
COM    ../src/hmmbuild -F rrm.hmm rrm.slx
COM    ../src/hmmcalibrate rrm.hmm
NSEQ   70
DATE   Wed Jul  8 08:13:25 1998
CKSUM  2768
GA      13.3 0.0
TC      13.40 0.60
NC      13.20 13.20
XT      -8455      -4 -1000 -1000 -8455      -4 -8455      -4
NULT     -4 -8455
NULE     595 -1558      85      338 -294      453 -1158      (...)
EVD     -49.999123 0.271164
HMM
      A      C      D      E      F      G      H      I (...)
      m->m  m->i  m->d  i->m  i->i  d->m  d->d  b->m  m->e
      -21      * -6129
1 -1234 -371 -8214 -7849 -5304 -8003 -7706 2384 (...) 1
- -149 -500 233 43 -381 399 106 -626 (...)
- -11 -11284 -12326 -894 -1115 -701 -1378 -21 *
2 -3634 -3460 -5973 -5340 3521 -2129 -4036 -831 (...) 2
- -149 -500 233 43 -381 399 106 -626 (...)
- -11 -11284 -12326 -894 -1115 -701 -1378 * *
```

(...)

Example output (from HMMER)

Alignments of top-scoring domains:

S13421: domain 6 of 8, from 928 to 1075: score 65.8, E = 1.5e-20

```
      *->vilealvnssShLSaeekalVkslWYgKVegnaeeiGaeaLgRlFvv
      +                LSa e a Vk++W   V+ ++ ++G  ++  lF +
S13421  928      G-----LSAREVAVVKQTW-NLVKPDLMGVGMRIFKSLFEA 962

      YPwTqryFphFgdLssldavkgspkvKaHGkKVltalgdavkhLDdtgnl
      +P  q+ Fp+F+d+ +ld +++ p v +H   V t l++ ++ LD   nl
S13421  963 FPAYQAVFPKFSDV-PLDKLEDTPAVGKHSISVTTKLDELIQTLDEPANL 1011

      kgalakLSelHadklrVDPeNFklLghvllvvvLaehfgkdftPevqAAwd
      +   +L+e H   lrV+   Fk +g+vlv L   +g f+  +  +w
S13421 1012 ALLARQLGEDH-IVLRVNKPMFKSFGKVLVRLLENDLGQRFSSFASRSWH 1060

      KflagvanaLahKYr<-*
      K++++++  +++
S13421 1061 KAYDVIVEYIEEGLQ      1075
```

Here, **target sequence S13421** is aligned with the states of a given profile HMM. The top sequence shows either an alignment to **match states** (UPPER CASE) or to **insert states** (lower case) for a single *representative sequence* in the family. A dash in the bottom sequence indicates use of a delete state.

How can we prevent over-fitting?

- **REGULARISATION**

- **Pseudo-counts**

- Add a constant c to the counts taken from alignment i.e. start counting at c rather than 0!

- **Prior Information (Dirichlet Mixtures)**

- Combines prior information regarding amino acid frequencies at each step of training
 - Can think of it as using variable “context sensitive” pseudocounts
 - Prior distribution used at each step depends on:
 - Number of examples seen in a given column
 - Distribution of examples seen

- **SEQUENCE WEIGHTING**

- Some sequences are more frequent than others
 - Your model should ideally not reflect this
 - Give the less frequent sequences more weight

- Get more data!

Sequence Weighting

- The simplest approach to sequence weighting is to weight each sequence by its average DISSIMILARITY to other sequences
- Calculate all pairwise dissimilarity scores between sequences (e.g. 100 - % Identity)
 - NOTE: Can convert similarity scores to dissimilarities by e.g. subtracting similarity scores from a constant
- Calculate mean dissimilarity for each sequence
- Can normalise average dissimilarity scores such that:

$$\sum \bar{d}_i = 1.0$$

Iterative Profile/HMM Generation

1. Close homologues of a *seed sequence* are first found using a rapid databank search technique (e.g. BLAST) and a suitably conservative E-value threshold.
2. These sequences are multiply aligned (either using an HMM approach or a classic multiple sequence alignment technique such as CLUSTALW).
3. An initial HMM is constructed based on this seed alignment (most often with Viterbi training for speed).
4. The HMM is used to identify new members of the family in the database according to the log-odds ratio above (i.e. sequences which score above a set threshold and which were not already found). If no new members are found then the process terminates.
5. A new alignment is generated by aligning the new members to a profile generated from the score of the HMM match states (see above).
6. A new HMM is generated based on this new alignment and the process repeats from Step 4 until no more new sequences are found.



HMMER

Biosequence analysis using profile hidden Markov Models

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protein sequence vs protein sequence database

[Paste a Sequence](#) | [Upload a File](#) | [Accession Search](#)

Paste in your sequence or use the [example](#) ?

>ARB Sequence

```
GVSGSCNIDVVCPEGDGRRDIIRAVGAYSKEGTLACTGSLVHNTANDRKMYFLTAHHCGMGTASTAASIVVYWHYQNSTCRAPNTPASGANGDGSMSQTQSGSTVKATYATSDFTLLELNNAANPAFNLFWAGW  
DRDQNYFGAIAIHHPNVAEKRI SNSTSTSTSTFVWGGGAGTTHLNVQWQPSGGVTEPGSSGSPFYSPEKRVLGQLHGGPSSCSATGTNRSDQYGRVFTSWTGGGAAASRLSDWLDPASTGAQFIDGLDS
```

[Submit](#)[Reset](#)

▼ Sequence Database ?

Frequently used databases:

[Reference Proteomes](#)[UniProtKB](#)[SwissProt](#)[PDB](#)[AlphaFold](#)[Ensembl](#)

Current database selection:

Reference Proteomes ▼



PHMMER Results

[Search Again](#)[Score](#)[Taxonomy](#)[Domain](#)[Download](#)

Sequence Features ?

[â° disorder](#) [â° coiled-coil](#) [â° tm & signal peptide](#)

Distribution of Significant Hits ?



■ Bacteria ■ Eukaryota ■ Archaea ■ Viruses ■ Unclassified Sequences ■ Other Sequences

Did you know? Clicking the button customise, in the table header below, gives you the opportunity toggle up to twelve columns of data in this table.

[hide this X](#)

Also, have a look at the new **Cross-references** column, showing references to other resources at the EBI.

« First « Previous **Page 1** of 5 Next » Last »

Significant Query Matches (456) in *uniprotrefprot* (v.2021_04)

[Customise](#)

	Target	Species	E-value
>	A0A0S2DKV3_LYSEN	Lysobacter enzymogenes	7.5e-167
>	A0A0S2FEY4_9GAMM	Lysobacter antibioticus	3.0e-164
>	A0A0S2FEZ7_9GAMM	Lysobacter antibioticus	1.4e-135
>	A0A0S2DKZ5_LYSEN	Lysobacter enzymogenes	6.3e-121
>	A0A3M8SXF8_9GAMM	Lysobacter psychrotolerans	9.0e-120
>	A0A2M9E9B2_9GAMM	Xanthomonadaceae bacterium NML91-0213	1.6e-116



HMMER

Biosequence analysis using profile hidden Markov Models

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protein alignment/profile-HMM vs protein sequence database

Using Existing Search as Input



This search will be run with an HMM generated from the **456** hits above threshold in your previous search [817118B4-D764-11EE-BA4A-486417302E6A.1](#).

▼ Sequence Database ?

Frequently used databases: [Reference Proteomes](#) [UniProtKB](#) [SwissProt](#) [PDB](#) [AlphaFold](#) [Ensembl](#)

Current database selection:

Ensembl Human ▼

▼ Restrict by Taxonomy ?

☒ Taxon search

Organism:

☐ Include all taxa

Human (taxid: 9606)



HMMSEARCH Results

[Search Again](#)[Score](#)[Taxonomy](#)[Domain](#)[Download](#)

Your search was restricted to include

Homo sapiens (taxid:9606)

Try the search again without restrictions. [Resubmit](#)

Distribution of Significant Hits ?



Did you know? Clicking the button customise, in the table header below, gives you the opportunity to toggle up to twelve columns of data in this table.

[hide this X](#)

Also, have a look at the new **Cross-references** column, showing references to other resources at the EBI.

Significant Query Matches (6) in *ensembl* (v.104)

[Customise](#)

	Target	Species	E-value
>	ENSG00000151006.7	Homo sapiens	1.2e-05
>	ENSG00000153802.12	Homo sapiens	0.00034
>	ENSG00000189057.11	Homo sapiens	0.00059
>	ENSG00000189057.11	Homo sapiens	0.00064
>	ENSG00000283539.2	Homo sapiens	0.00097
>	ENSG00000283539.2	Homo sapiens	0.002



HMMSEARCH Results

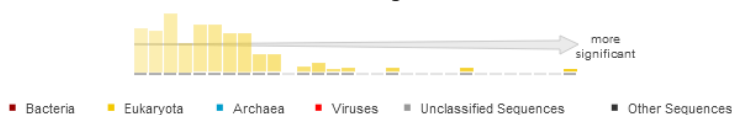
[Search Again](#)[Score](#)[Taxonomy](#)[Domain](#)[Download](#)

Your search was restricted to include

Homo sapiens (taxid:9606)

Try the search again without restrictions. [Resubmit](#)

Distribution of Significant Hits



Did you know? Clicking the button customise, in the table header below, gives you the opportunity toggle up to twelve columns of data in this table.

[hide this X](#)

Also, have a look at the new **Cross-references** column, showing references to other resources at the EBI.

« First « Previous **Page 1 of 4** Next » Last »

Significant Query Matches (291) in *ensembl* (v.104)

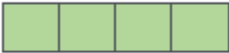
[Customise](#)

	Target	Species	E-value
>	ENSG00000153802.12	Homo sapiens	3.2e-104
>	ENSG00000178297.14	Homo sapiens	3.3e-78
>	ENSG00000178297.14	Homo sapiens	3.9e-78
>	ENSG00000151006.7	Homo sapiens	1.4e-60
>	ENSG00000283539.2	Homo sapiens	2.0e-60

Language Modelling in Biology

- Techniques for natural language processing (NLP) can also be applied to biological sequences
- Language models now exist for:
 - Protein sequences
 - Protein structures
 - DNA sequences
 - RNA sequences
 - Textual annotations from data banks

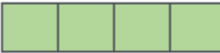
Embeddings

x_1 

The

Il

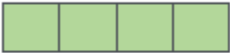
Threonine

x_2 

Black

Gatto

Glycine

x_3 

Cat

Nero

Leucine

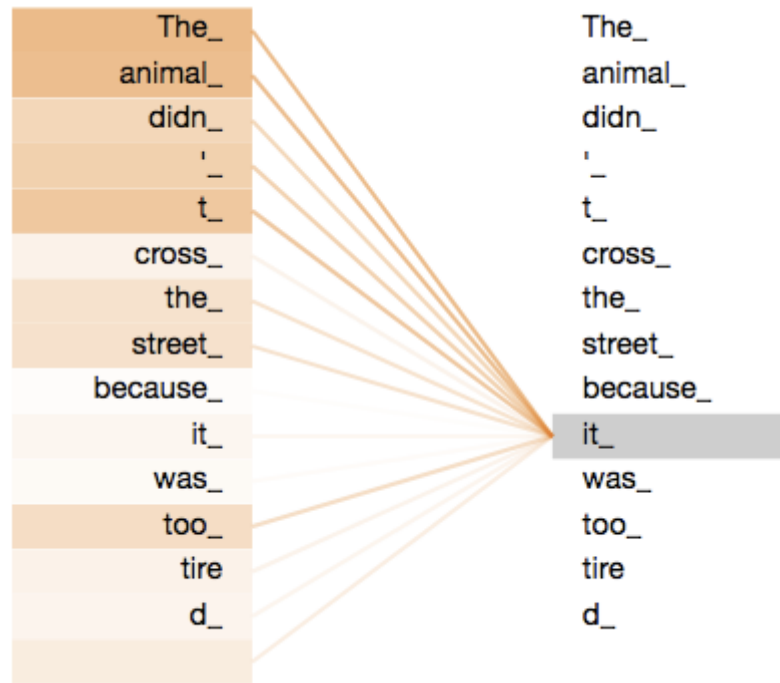
Transformers

- The goal of a transformer is to take a set of vectors representing something (words/letters/amino acids) and TRANSFORM then into a new set of vectors which have extra contextual information added
- These are sequence-to-sequence methods i.e. inputs are strings of vectors e.g. a sentence or a protein sequence
- They treat data as sets and so are PERMUTATION INVARIANT:
 - $[\text{Embed}(A), \text{Embed}(B), \text{Embed}(C)] = [\text{Embed}(C), \text{Embed}(B), \text{Embed}(A)]$

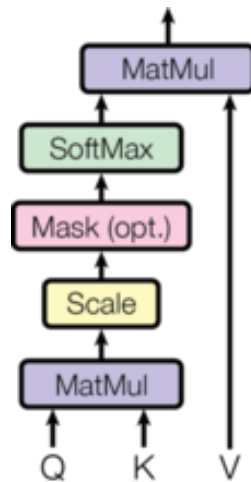
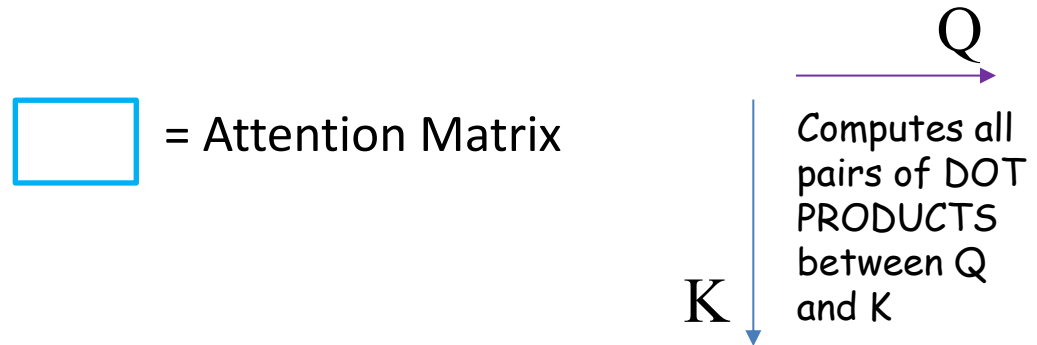
Key reference: Attention is all you need. Vaswani et al. 2017

<https://papers.nips.cc/paper/2017>

(Self-)Attention



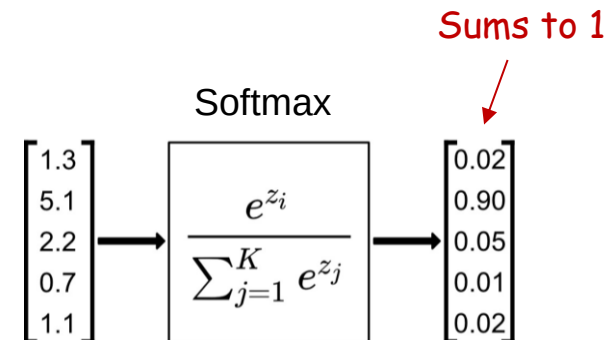
How attention can be computed (Scaled Dot-product Attention)



$$\text{softmax}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right)$$

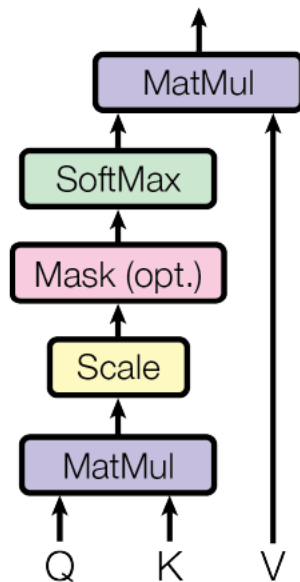
Z

Q = Queries K = Keys V = Values



Transformer Models

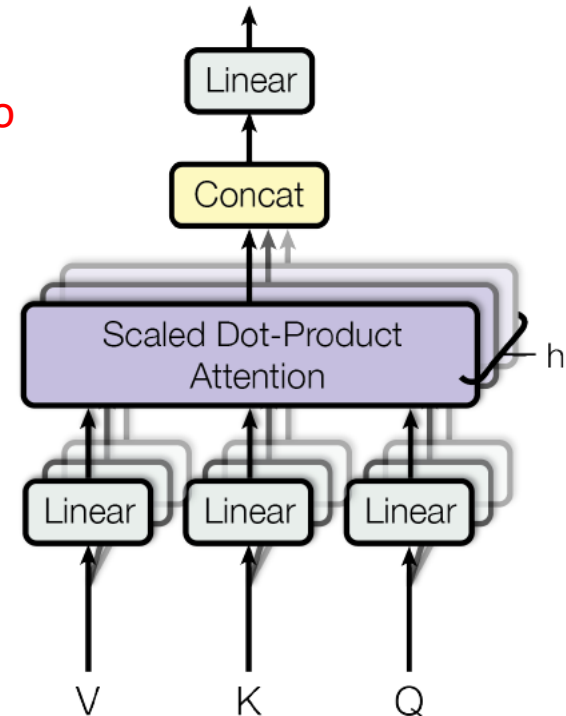
Scaled Dot-Product Attention



These are
concatenated to
make the final
output

Each “head”
produces its
own (QK)V

Multi-Head Attention



Bidirectional Encoder Representations from Transformers (BERT)

- How can we train a transformer to produce useful embeddings?
- We need an objective function we can optimize!

I took a bite from a ??? green apple.

I ??? a ??? from a juicy ??? apple.

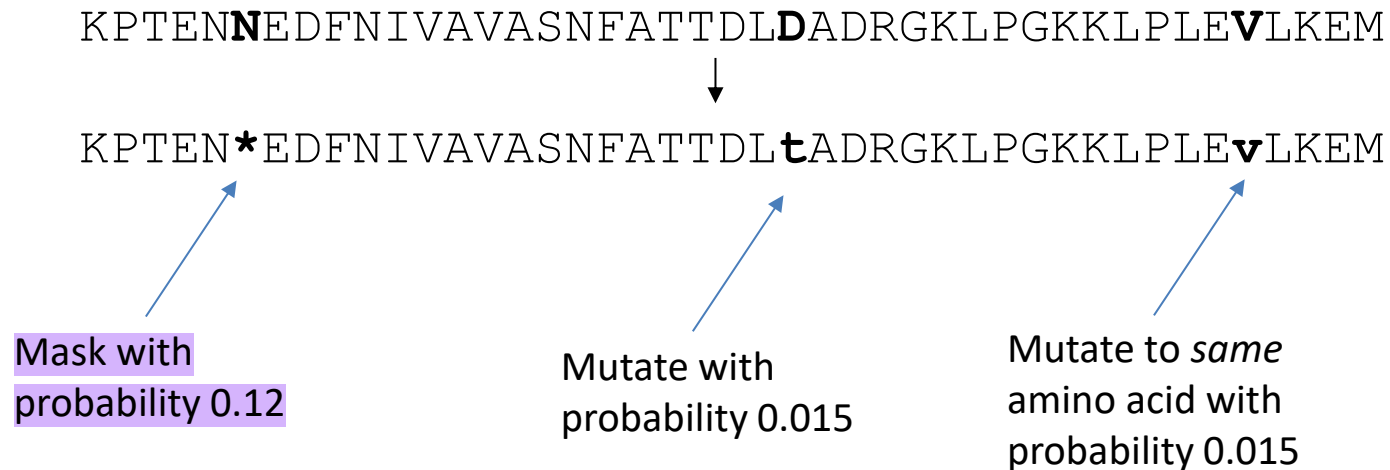
??? took a bite ??? a juicy green ???.

BERT Loss

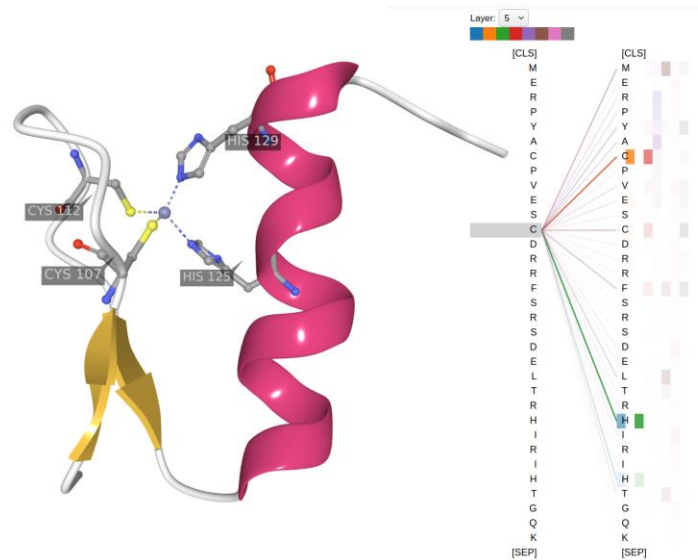
- The transformer encoder is used to encode randomly *masked* versions of the same text
- This just means replacing the original word/letter/amino acid with a placeholder which means “don’t know”
- The transformer is trained to correctly predict the tokens that have been masked out
- This is usually scored with a cross-entropy loss function
- This is supervised training but *unsupervised* with respect to any downstream tasks

BERT for protein sequences

ESM pre-training scheme...



BERT with amino acid sequences... Protein Language Models (PLMs)



Looking at the attention maps (softmax outputs) after BERT training, can indicate which amino acid pairs are functionally linked or even in contact!

Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences

✉ Alexander Rives, Joshua Meier, ✉ Tom Sercu, Siddharth Goyal, Zeming Lin, Jason Liu, Demi ...
+ See all authors and affiliations

PNAS April 13, 2021 118 (15) e2016239118; <https://doi.org/10.1073/pnas.2016239118>

Edited by David T. Jones, University College London, London, United Kingdom, and accepted by Editorial Board Member William H. Press December 16, 2020 (received for review August 6, 2020)

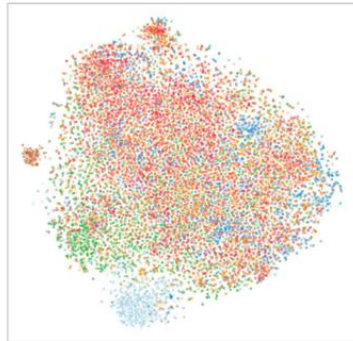
ProtTrans: Towards Cracking the Language of Lifes Code Through Self-Supervised Deep Learning and High Performance Computing.

Elnaggar A, Heinzinger M, Dallago C, Rehawi G, Wang Y, Jones L, Gibbs T, Feher T, Angerer C, Steinegger M, Bhowmik D, Rost B

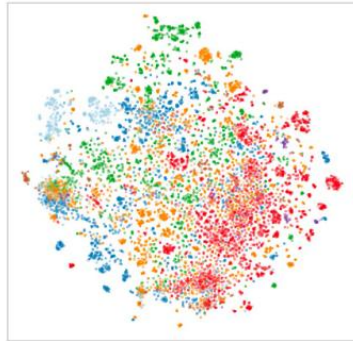
IEEE Transactions on Pattern Analysis and Machine Intelligence, 07 Jul 2021, PP

DOI: 10.1109/tpami.2021.3095381 PMID: 34232869

Secondary structure prediction using sequence embeddings



Untrained



Trained

● All alpha ● Multi-domain
● All beta ● Membrane, cell surface
● Alpha & beta (a|b) ● Small proteins
● Alpha & beta (a+b)

B Structure: SCOPe

Accuracy (Q_3) scores

<i>Dataset</i>	<i>CASP12</i>	<i>NEW364</i>
DeepProtVec	62.9	64.7
ProtTXL*	71.5	72.8
ProtTXL-BFD*	71.7	72.2
DeepSeqVec	73.0	76.0
ProtXLNet*	73.7	77.3
ProtElectra*	73.9	78.1
ProtAlbert*	74.6	78.5
ProtBert*	75.0	80.1
ProtBert-BFD*	75.8	81.1
ESM-1b	76.9	82.6
ProtT5-XXL-BFD*	77.7	81.6
ProtT5-XL-BFD*	77.5	82.0
ProtT5-XXL-U50*	79.2	83.3
ProtT5-XL-U50*	81.4	84.8
NetSurfP-2.0	82.0	84.3

Different ways of using language models (LMs) in bioinformatics

- Single-task supervised training
 - Train a transformer LM on labelled sequences to predict new labels.
- Unsupervised pretraining + supervised fine-tuning
 - Train a transformer LM using BERT on unlabelled sequences. Then add a new output layer and continue training it on a small number of labelled sequences to predict new labels.
- Unsupervised pretraining + supervised training of small downstream classifier
 - Train a transformer LM using BERT on unlabelled sequences, then freeze the weights. Use the frozen LM outputs on labelled sequences to generate inputs to train a new model to predict new labels.
- FUTURE: Unsupervised pretraining at scale + prompting (few-shot learning)
 - Train a very large transformer LM autoregressively on very diverse sequence + text data. Then try to find suitable prompts to induce it to predict labels.

Scaling laws for pLMs



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bioRxiv posts many COVID19-related papers. A reminder: they have not been formally peer-reviewed and should not guide health-related behavior or be reported in the press as conclusive.

New Results

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Evolutionary-scale prediction of atomic level protein structure with a language model

Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, Alexander Rives

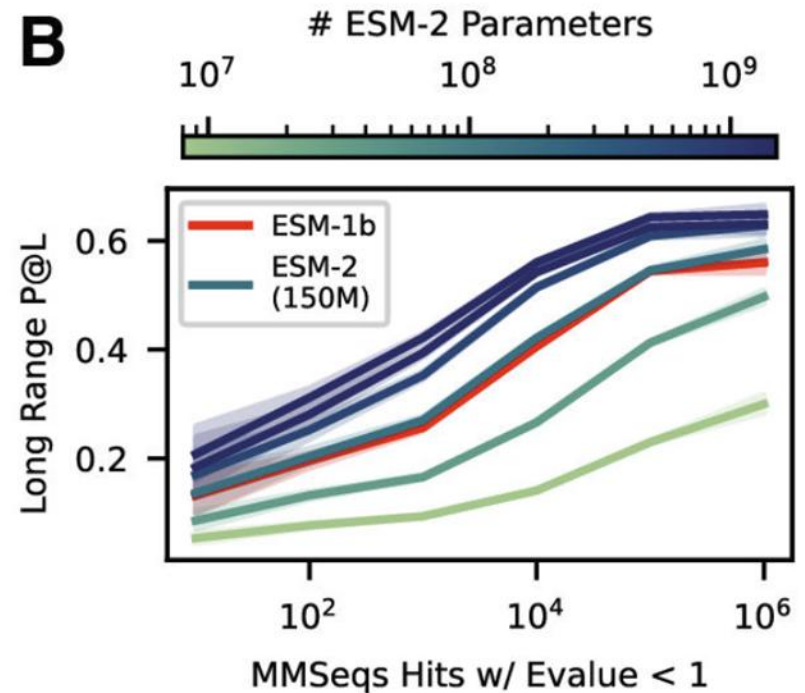
doi: <https://doi.org/10.1101/2022.07.20.500902>

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Posted December 21, 2022.

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[Revision Summary](#)

B



Protein Language Models (pLMs)

- pLMs are an active area of research
- Currently they perform less well than the best alignment-based methods in many applications
 - BERT training does not sample indels sufficiently?
 - Positional encoding methods are poor?
 - Insufficient training data?
- However, they obviate the need for alignments to be calculated and so offer significant efficiency benefits